

HOW TO OPTIMIZE A QUANTUM OPTIMIZATION ALGORITHM

Integration of Dynamic ADAPT-QAOA and QuCLEAR to reduce quantum gates in QAOA Circuits

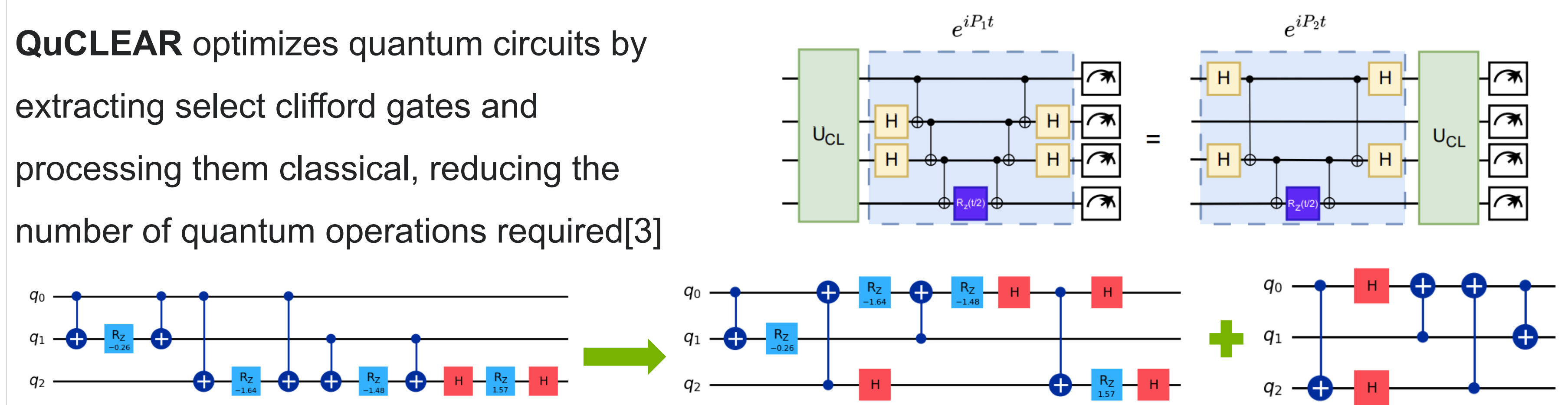
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BACKGROUND

- The Quantum Approximation Optimization Algorithm (QAOA) solves QUBO problems like MaxCut by alternating between the Cost Hamiltonian H_c , which encodes the problem, and the Mixer Hamiltonian H_m which perturbs the system and lets us explore the solution space. These Hamiltonians use parameterized quantum gates which are optimized classically to minimize H_c
- A **key limitation** of QAOA is the high number of quantum gates required to encode the Hamiltonians[1]
- QuCLEAR and Dynamic ADAPT-QAOA (AQ) are unique optimization techniques that reduce the number of quantum gates needed for Standard QAOA

QuCLEAR: QUANTUM CLIFFORD EXTRACTION AND ABSORPTION

QuCLEAR optimizes quantum circuits by extracting select clifford gates and processing them classical, reducing the number of quantum operations required[3]



Dynamic ADAPT-QAOA

Dynamic AQ selects the mixer that will maximize the gradient, and determines is the H_c for a given layer is needed, optimizing the circuit by reducing the number of quantum gates required for convergence[2]

The energy variation due to the added parameters

$$\delta E_n(\beta_n, \gamma_n; A) = \langle \Psi_n(\beta_n, \gamma_n; A) | H | \Psi_n(\beta_n, \gamma_n; A) \rangle$$

Enables the definition of the energy gradient

$$\mathcal{G}_p(\gamma_n; A) \equiv \frac{\partial}{\partial \beta_n} \delta E_n(\beta_n, \gamma_n; A) \Big|_{\beta_n=0}$$

Evaluating the gradient identifies the optimal mixer

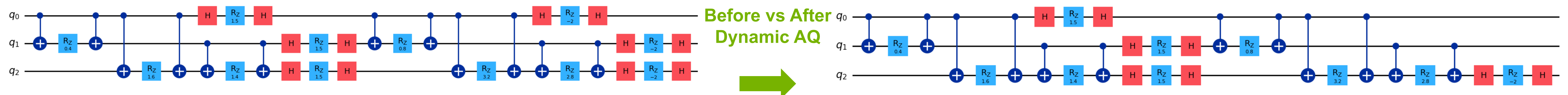
$$M_n = \operatorname{argmax}_{A \in n} [\mathcal{G}_n(\gamma_n; A)]$$

MOTIVATION

- Will combining optimizations reduces quantum gates and error?
- Can Dynamic AQ and QuCLEAR work in tandem?
- Does the order of optimization matter?
- Is this a scalable solution?

METHODS

- Proof that order of optimization is inconsequential
- Empirical analysis of clifford gate counts for QAOA, Dynamic AQ, QuCLEAR, and their combination
- A look at scalability of both approaches combined



ORDER OF OPTIMIZATION

Initial setup: Layers of alternating H_c and H_m

Layer 0 = $|0\rangle$

Layer 0 + 1 = $M_1 C_1 |0\rangle$

$\sum_{0}^{n-1} \text{Layers} = M_{n-1} C_{n-1} \dots M_1 C_1 |0\rangle$

First: Use Dynamic AQ to determine the mixer M_n for layer n

$$M_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle \psi_{n-1} | e^{-i\beta_n A} e^{-i\gamma_n H} | H | e^{-i\beta_n A} e^{-i\gamma_n H} \psi_{n-1} \rangle \right| \right]$$

$\sum_{0}^n \text{Layers} = M_n C_n M_{n-1} C_{n-1} \dots M_1 C_1 |0\rangle$

Second: Use QuCLEAR to determine the optimized circuit and clifford circuit

$$\sum_{0}^n \text{Layers} = M_n C_n U_{CL} U_{\text{opt}} |0\rangle = U_{CL} M'_n C'_n U_{\text{opt}} |0\rangle$$

where $M'_n = U_{CL}^\dagger M_n U_{CL}$ $C'_n = U_{CL}^\dagger C_n U_{CL}$

Now: Reverse order and use QuCLEAR first to optimize layers 0 to $n-1$

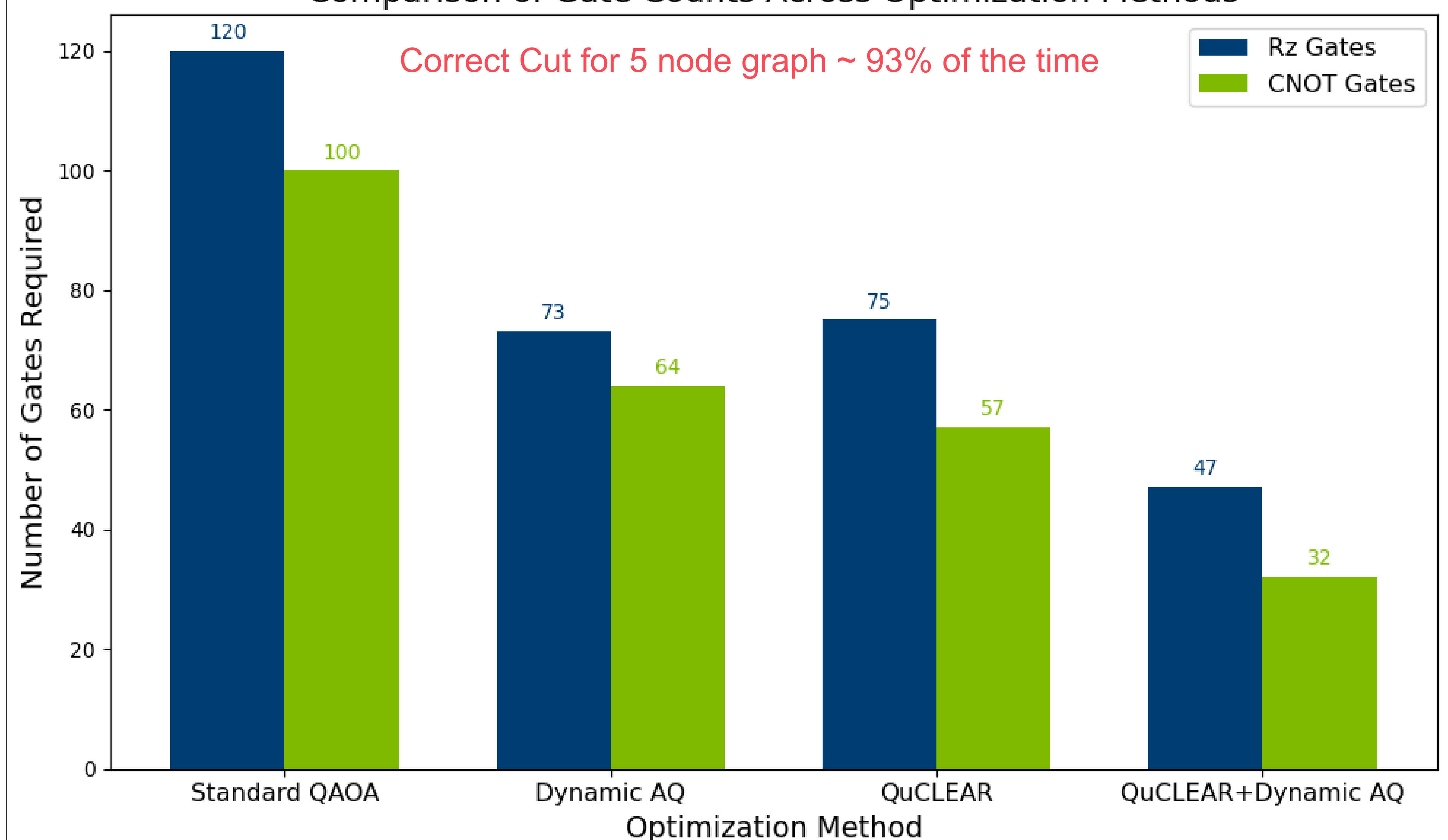
$$\sum_{0}^{n-1} \text{Layers} = U_{CL} U_{\text{opt}} |0\rangle$$

Next: Use the QuCLEAR optimized layers and apply Dynamic AQ to get M_n

$$M_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle 0 | U_{\text{opt}}^\dagger U_{CL}^\dagger e^{-i\beta_n A} e^{-i\gamma_n H} | H | e^{-i\beta_n A} e^{-i\gamma_n H} U_{CL} U_{\text{opt}} | 0 \rangle \right| \right]$$
$$M'_n = \operatorname{argmax} \left[\left| \frac{\partial}{\partial \beta} \langle 0 | U_{\text{opt}}^\dagger e^{-i\beta_n A'} e^{-i\gamma_n H'} U_{CL}^\dagger | H | U_{CL} e^{-i\beta_n A'} e^{-i\gamma_n H'} U_{\text{opt}} | 0 \rangle \right| \right]$$
$$\sum_{0}^n \text{Layers} = U_{CL} M'_n C'_n U_{\text{opt}} |0\rangle$$

EMPIRICAL RESULTS

Comparison of Gate Counts Across Optimization Methods

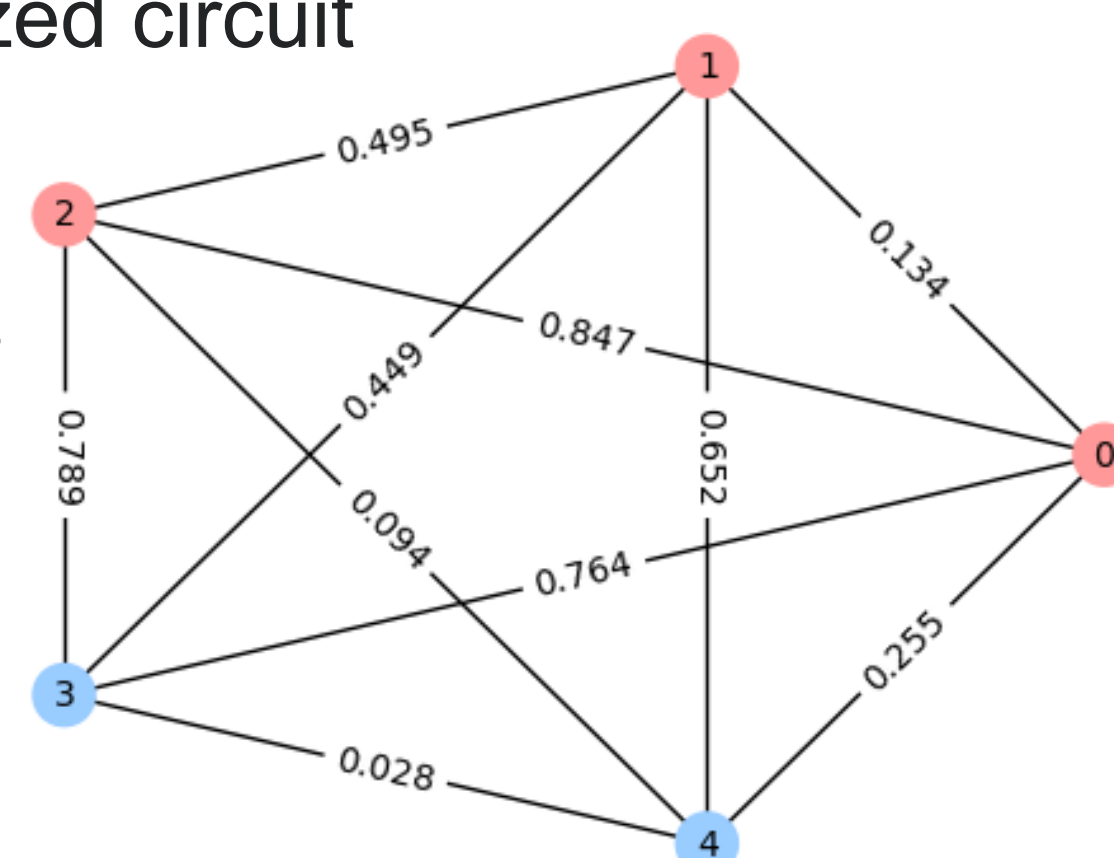


SCALABILITY

- In a 5-node example, QuCLEAR cut CNOT gates by 43% and RZ gates by 37.5%, while Dynamic AQ reduced them by 36% and 39%, respectively. Combined, they achieved reductions of 68% in CNOT gates and 60% in RZ gates
- But Dynamic AQ's gradient calculation introduces a computational bottleneck due to expensive matrix multiplications
- Inspired by Adapt-VQE, this is solved by calculating gradients in $O(n)$ time using FT-QC. Until such devices exist, we rely on costly classical methods for validation[4]

SUMMARY

- Combining Dynamic AQ and QuCLEAR reduces quantum gates and sources of error
- The application order of techniques does not affect the final optimized circuit
- Dynamic AQ scales well with quantum gradient computations



FUTURE RESEARCH

Dynamic AQ's costliest step is gradient calculation. Future work could shrink the mixer pool or use machine learning to predict the best mixer, avoiding gradient evaluations

REFERENCES

[1]Mazumder, A., & Tayur, S. (2024). Five starter problems: Solving quadratic unconstrained binary optimization models on quantum computers. *arXiv:2401.08989*. [2]Yanakev, N., et al. (2023). Dynamic-ADAPT-QAOA: An algorithm with shallow and noise-resilient circuits. *arXiv:2309.00047*. [3]Liu, J., et al. (2024). QuCLEAR: Clifford extraction and absorption for significant reduction in quantum circuit size. *arXiv:2408.13316*. [4]Zhu, L. An adaptive quantum approximate optimization algorithm for solving combinatorial problems on a quantum computer. *arXiv:2005.10258* (2022)

Abstract

Quantum computing holds great promise for addressing practical challenges. However, current quantum devices suffer from noisy quantum gates, which degrade the fidelity of quantum circuits. As a result, optimizing quantum circuits is essential for achieving useful results. In this paper, we explore the combination of two frameworks designed to optimize quantum circuits: QuCLEAR and Dynamic ADAPT-QAOA. First, we provide an overview of both frameworks, detailing their functions, key differences, and the individual benefits of each. Second, we present a mathematical proof that combining these two methods results in a more optimized circuit, with the order of application being inconsequential. Our empirical results for a 5-node example show that QuCLEAR reduces the CNOT gates by 43% and the RZ gates by 37.5%, while Dynamic ADAPT-QAOA achieves reductions of 36% and 39%, respectively. When combined, these methods reduce the CNOT gates by 68% and the RZ gates by 60%. Third, we compare the performance of standard QAOA with each approach individually and in tandem. Finally, we conclude with a discussion on scalability, interpretation of the results, and potential future work.